

Decision Sufficiency, Delayed Choice, and Witness-Bound Recovery

QIK-VRT: formal boundaries, finite verification, and provenance-bound continuation

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Abstract

This paper gives a minimal mathematical criterion for deterministic recovery under partial observability and applies it to two QIK-VRT contexts: the logical boundary around delayed choice/measurement independence and the Authority–Mirror–Witness recovery architecture. The core theorem is that a deterministic correct action C can factor through an observation Obs' exactly on those reachable observations whose fibers do not mix histories requiring different actions. Equivalently, $\ker(Obs') \subseteq \ker(C)$. A witness is therefore best characterized as an observation refinement that makes this kernel inclusion true, not intrinsically as a third replica or majority voter. The document also records an executed exhaustive finite-model check for the bounded Authority–Mirror fault model. Physical non-superdeterminism, SHA-256 collision resistance, arbitrary multi-fault tolerance, chip implementation, patent novelty, and completion-state claims are explicitly outside the established scope.

1 Minimal decision-sufficiency theorem

Let $\mathcal{H}_{\text{adm}}$ be the admissible history space, $C : \mathcal{H}_{\text{adm}} \rightarrow \mathcal{A}$ the correct continuation action, and $Obs' : \mathcal{H}_{\text{adm}} \rightarrow \mathcal{O}$ the available observation. Define recovery-relevant equivalence by

$$H_1 \sim_R H_2 \iff C(H_1) = C(H_2).$$

For observation o , let

$$K(o) = \{H \in \mathcal{H}_{\text{adm}} \mid Obs'(H) = o\}.$$

The exact recovery criterion is

$$\boxed{RECOVER(o) \iff K(o) \neq \emptyset \wedge |\{C(H) \mid H \in K(o)\}| = 1.}$$

Otherwise the system must fail closed:

$$\boxed{K(o) = \emptyset \vee |\{C(H) \mid H \in K(o)\}| > 1 \Rightarrow FAIL_CLOSED.}$$

A unique past is unnecessary. What matters is uniqueness of the correct continuation:

$$Recover(o) = a \iff \forall H \in K(o) : C(H) = a.$$

2 Kernel normal form and factorization

The information-sufficiency theorem is

$$\boxed{C = S \circ Obs_W}$$

on the admissible history space if and only if C is constant on each reachable fiber:

$$\boxed{Obs_W(H_1) = Obs_W(H_2) \Rightarrow C(H_1) = C(H_2).}$$

Equivalently,

$$\boxed{\ker(Obs_W) \subseteq \ker(C).}$$

The selector is unique on the reachable image $Obs_W(\mathcal{H}_{\text{adm}})$; outside that image it may be defined arbitrarily or fail closed.

Impossibility

If admissible histories H_1, H_2 satisfy

$$Obs(H_1) = Obs(H_2), \quad C(H_1) \neq C(H_2),$$

then no universally correct deterministic selector S can satisfy $C = S \circ Obs$. The selector receives the same input in both histories and must return the same output, contradicting the different required actions. Thus

$$\boxed{\ker(Obs) \not\subseteq \ker(C) \Rightarrow \text{no universally correct deterministic recovery selector.}}$$

Sufficiency

If every nonempty reachable observation fiber is action-constant, define $S(o)$ as the unique action appearing in that fiber. Then S is well-defined on the reachable observation image and $C = S \circ Obs'$. Any two such selectors agree on that image.

3 Witness interpretation

A witness is not essentially a third copy or a majority vote. It is any refinement Obs_W that adds enough information to make

$$\boxed{\ker(Obs_W) \subseteq \ker(C).}$$

This yields the epistemic normal form:

Evidence is decision-sufficient exactly when its observation fibers do not mix the equivalence classes relevant to the correct action.

4 Delayed choice and the superdeterminism boundary

In a two-wing delayed-choice/Bell-style setting let λ be hidden/preparation state and a, b the settings. A clean ordering is

- t_0 : preparation,
- t_1 : local propagation,
- t_2 : spacelike-separated settings a, b ,
- t_3 : local outcomes $x = A(\lambda, a)$, $y = B(\lambda, b)$,
- t_4 : later classical comparison of $P(x, y \mid a, b)$.

The later comparison changes the evidential classification, not the earlier physical history:

$$\boxed{\text{later verification} \neq \text{retroactive physical mutation.}}$$

Measurement independence is represented at support level by

$$J(\lambda, a, b) \iff H(\lambda) \wedge S(a, b),$$

conceptually corresponding to $P(\lambda \mid a, b) = P(\lambda)$. For the formal candidate used in the Lean tranche,

$$\boxed{MI(M) \Rightarrow \neg SD_{MI}(M).}$$

However local response structure alone does not imply measurement independence: a finite common-cause model can keep each wing locally parameter-insensitive while correlating the allowed settings with λ . Hence the non-circular physical obligation remains

$$\boxed{\text{primitive QCE principles} \stackrel{?}{\implies} MI.}$$

5 Authority–Mirror–Witness specialization

The hardware architecture uses two non-volatile replica images A_n, M_n and an independent commit witness. A transaction follows

PREPARE $\rightarrow$ CROSS-VERIFY $\rightarrow$ COMMIT $\rightarrow$ ACKNOWLEDGE.

A conceptual witness is

$$W_n = H(n, H(A_n), H(M_n), W_{n-1}).$$

Without a witness, two locally valid divergent images may correspond to different hidden histories requiring different recovery actions. Their observation fiber therefore mixes recovery-relevant equivalence classes and deterministic selection is impossible. With a valid witness, the observation is refined until a unique certified epoch/action remains. This is the precise sense in which the system implements

self-healing $\wedge$ no self-confirmation.

6 Executed finite-model verification

A separate executable checker exhaustively enumerated the bounded Authority–Mirror model. It returned

FINITE_MODEL_EXHAUSTIVE_CHECK_PASSED.

The executed case counts were:

Case family	Cases
Crash recovery	945
Effect-ACK generation	945
Recovery idempotence	945
Single-replica reconstruction	756
Four-step payload sequences	81
Hidden-history impossibility cases	36
Witnessless ambiguity cases	18

This is exhaustive only for the explicitly bounded state/fault model.

7 Lean relation layer

The repository materializes three relevant Lean tranches:

- ‘QuantumFoundations/MasurementIndependence.lean’: conditional exclusion of the defined measurement-dependent superdeterministic candidate plus a finite countermodel to the invalid locality $\Rightarrow$ independence inference;
- ‘Hardware/AuthorityMirrorWitness.lean’: T01–T22 for witness-bound recovery and Effect ACK within the abstract hardware model;
- ‘Decision/ObservationSufficiency.lean’: recovery equivalence, kernel inclusion, mixed-fiber impossibility, factorization on the reachable image, uniqueness there, and observation-refinement sufficiency.

The exact-head axiom audit includes all three tranches. A successful execution-bound Lean 4.19 kernel receipt for the current combined head has not been observed in this environment; source materialization is therefore not promoted to kernel-execution success.

8 Publication and protocol boundary

Zenodo publication is fail-closed pending final exact-byte artifacts, a successful execution-bound machine-proof/kernel receipt, repository authorization and credential gates, and post-publication byte verification. No normative Effect-Acknowledgement wire-format, state-machine, DONE-predicate, security, or interoperability change is introduced here. Therefore the current IETF disposition is ‘NO_PROTOCOL_CHANGE_REQUIRED’.

9 Established and non-established claims

Established in the stated mathematical or finite scope:

- mixed observation fibers make universal deterministic recovery impossible;
- action-constant reachable fibers are sufficient for factorization and unique selection on the reachable image;
- a witness can be characterized as an observation refinement satisfying kernel inclusion;
- measurement independence conditionally excludes the defined measurement-dependent superdeterministic candidate;
- local two-wing response structure alone is insufficient to establish measurement independence;
- the bounded hardware checker passed every enumerated case listed above.

Not established by these results:

- that nature satisfies measurement independence or is non-superdeterministic;
- SHA-256 collision resistance as a theorem;
- arbitrary multi-fault tolerance or gate-level timing correctness;
- physical chip implementation or patent novelty/inventive step;
- ‘PASS’, ‘FINAL_PASS’, or ‘EFFECT_ACK_DONE’.

10 Conclusion

The most compact statement is

$$\boxed{\ker(Obs') \subseteq \ker(C)}.$$

We do not need to know which past occurred uniquely. We need sufficient evidence that every still-admissible past compatible with what we observe requires the same correct continuation. That criterion is the general information-theoretic core behind witness-bound recovery and, more broadly, deterministic decision making under partial observability.

Quod erat demonstrandum – for the explicitly delimited mathematical and finite-model claims.
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